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SECTION 1 · EXECUTIVE INTRODUCTION

Maheshchandra Hegde

Founder & CTO | Building AI-Driven 3D Asset Platforms & Spatial Computing Solutions
[@srushtilabs.com](https://srushtilabs.com)

Innovative technologist and product strategist with 18+ years of experience architecting end-to-end digital products, interactive 3D experiences, and domain-specific SaaS platforms. Co-Founder and Technology Leader at Technoyana Digital Transformation Services Pvt. Ltd., driving product architecture and creative tech vision across flagship divisions:

- Twitan.com — AI-driven sports management app & operational SaaS suite (Shutlify badminton OS).
- Srushtilabs.com — Generative AI & modular 3D computing platform for games, simulations, and 3D visualization.

Holds a postgraduate degree in Human Interface Design & Development (M.S. in Computing, Robert Gordon University, UK) and an engineering background in Electronics & Communication (B.E.). Expertise spans modern web/mobile stacks (React, TypeScript, Angular, Web Components, Node.js, Firebase), interactive 3D/PBR pipelines, and design-to-code automation.

Architectural Heritage: Growing up assisting my architect father in his studio provided my earliest practical foundation in architectural drawings, CAD drafting, and spatial visualization—cultivating a natural synergy between software systems and 3D engineering.

Technoyana & SrushtiLabs: Generative 3D Platforms & Spatial Computing

Voxelforge AI

Modular 3D Generative AI Platform | srushtilabs.com/voxelforge/

- **Role & Entity:**

Founder & CTO at SrushtiLabs (Technoyana Digital Transformation Services Pvt. Ltd.).

- **Core Product Innovation:**

Developed Voxelforge AI, a generative AI product creating modular 3D assets based on user prompt inputs.

- **Multi-Platform Assembly:**

Produces game-ready, low-poly 3D models and bundles optimized for immediate integration into Unreal Engine, Unity, and real-time WebGL engines.

- **Design-to-Asset Speed:**

Automates prompt-to-3D geometry conversion, texture baking, and polygon optimization, reducing prototyping turnaround from days to minutes.

- **Live Demonstration:**

Fully functional application deployed and accessible at srushtilabs.com/voxelforge/.

Ayam3d (Ayam = Dimension)

Parametric AI 3D Model Generation | ayam3d.in

- **Concept & Vision:**

Extended the Voxelforge concept into Ayam3d ('Ayam' meaning Dimension), focusing on parametric-based 3D model generation from natural language prompts.

- **Instant Generation:**

Emphasizes generating structured 3D models in seconds utilizing custom trained AI models and geometric constraint solvers.

- **Investor Interest:**

Successfully generated early investor interest, validating the commercial potential of real-time generative 3D synthesis for digital twins, architecture, and games.

- **Active Development:**

Currently under active development; seeking strategic industry partners and promoters to scale the underlying AI pipeline.

- **Application Reference:**

Preliminary landing page and concept documentation available at ayam3d.in.

Cyient / PHILIPS: Intellispace Critical Care & Anesthesia (ICCA)

Philips ICCA

Intellispace Critical Care & Anesthesia

Role:

Senior Technology Leader (React / NodeJS)

Tenure:

Nov 2023 – Mar 2025

Client & Facility:

Cyient Limited for Philips Healthcare, Bangalore

Operational Reality:

Mission-critical hospital ICU and anesthesia suites operating continuously 24/7.

Delivery Mandate:

Delivered clinical software passing all stringent quality, reliability, and security gates established by Philips and hospital networks.

1. Rigorous Quality Benchmarks & Zero-Downtime Reliability

Successfully delivered the software to meet demanding medical quality requirements set by Philips and healthcare providers. Architected fail-safe clinical data rendering with zero memory leaks, ensuring bedside patient monitoring displays remain completely responsive across multi-day uninterrupted operations.

2. Comprehensive Security Architecture & Policy Enforcement

Enforced strict enterprise cybersecurity measures as per Philips global policies: integrated Multi-Factor Authentication (MFA), protected against Cross-Site Scripting (XSS) and injection vulnerabilities, enforced role-based access control (RBAC), and maintained immutable audit logs for sensitive patient medical data.

3. End-to-End Best Practices & Scalable Modular Delivery

Led the end-to-end technical execution—from architectural design and modular component breakdown to automated test suites and production deployment. Delivered a fully supported software product backed by complete documentation and service support protocols.

Ness Digital Engineering & Moonraft Innovation Labs (UST Global Unit)

Ness Digital Engineering

Senior Analyst / Tech Lead | Oct 2019 – Apr 2020

- **Project Focus:**
Entertainment Management Application for a major client.
- **Technology Stack:**
Angular, TypeScript, AWS cloud infrastructure, Node.js.
- **System Architecture:**
Architected the complete single-page application structure, establishing clean state management and secure API communication layers.
- **Standards & Best Practices:**
Enforced accessibility standards (WCAG), code quality linters, and strict frontend security guidelines across the engineering team.
- **Technical Leadership:**
Led the development squad through agile sprints, code reviews, and end-to-end technical deployment to client environments on schedule.

Moonraft Innovation Labs

UI Architect | May 2019 – Jul 2019 (UST Global Specialized Unit)

- **Enterprise Design System:**
Architected and engineered a centralized cross-framework UI Design System using LitElement Web Components, React, AngularJS, Node.js, and npm packaging.
- **Eliminated Duplication:**
Stored all standardized UI components in a central repository, preventing teams from reinventing the wheel for new client projects.
- **Unified Look & Feel:**
Guaranteed visual consistency and brand coherence across all digital products from a single, shared codebase.
- **Automated Upgradability:**
When the core component library is updated, client projects automatically inherit updates and enhancements via standard package management.
- **Luxury Hotel Mobile App:**
Led mobile application development for a premier Indian hotel group (TLC Group of Hotels) built using Ionic and Angular.

UST Global / CISCO: Stadium Vision Director Migration

Cisco Stadium Vision

Stadium Vision Director Enterprise Platform

Role:

Senior Analyst & Associate Project Manager

Tenure:

Feb 2015 – Feb 2019 (4+ Years Dedicated Tenure)

Client:

Cisco Systems India Private Limited via UST Global

Product Scope:

Centralized digital media, live video distribution, and dynamic stadium display management deployed at major international sporting arenas.

Core Mandate:

Modernize a massive legacy application suite with numerous active sub-applications and active global customers.

1. Phased Zero-Downtime Migration from Legacy Flash/Flex

The platform was originally built using Adobe Flash, Flex, and ActionScript. Took technical leadership in architecting and executing a phased migration to modern Angular and React modules without interrupting active stadium operations or breaking existing client workflows.

2. Hybrid Interoperability & Multi-Stack Engineering

Maintained seamless interoperability between legacy components and modernized Single Page Application (SPA) modules using TypeScript, Angular, React, and Node.js backend services until complete deprecation was achieved.

3. Sub-Application Management & Sustained Excellence

Governed multiple parallel UI sub-applications serving venue operators, technicians, and broadcast controllers. Awarded multiple Certificates of Excellence by UST Global (2015, 2016, 2018) for client-first commitment and outstanding delivery.

ThoughtFocus Technologies: Aftermarket Parts 3D Explorer & Enterprise Solutions

Aftermarket Parts Explorer

UI Architect & Tech Lead | Oct 2011 – Mar 2014

- **Platform Scope:**

Massive parts marketplace serving aftermarket automotive & industrial parts vendors and buyers.

- **Scale & Catalog:**

Supported millions of parts across hundreds of manufacturer catalogs with deep search capabilities.

- **Interactive 3D Viewer:**

Architected an interactive 3D interface allowing buyers to inspect 3D models and dimensions of thousands of components before purchase.

- **Full E-Commerce Flow:**

Integrated shopping cart, quotation workflows, and secure payment gateway integrations.

- **Technologies Used:**

Adobe Flex, ActionScript, Mate architectural framework, HTML, CSS, and backend data services.

Enterprise UI Solutions

Prototyping & Multi-Client Solutions Delivery

- **Design-to-Code Pipeline:**

Established an early user involvement process—validating concepts with interactive mockups and prototypes prior to delivery.

- **Enterprise Search App:**

Guided team to design and bootstrap a high-speed enterprise search application with advanced filtering and metadata indexing.

- **Dairy / Milk Corporation App:**

Led the architecture and delivery of an operational management application for a large milk corporation handling distribution and supply logistics.

- **Technology Stack:**

HTML, CSS, JavaScript, TypeScript, Adobe Flex, Flash, and early Single-Page Application patterns.

- **Mentorship & Guidance:**

Coached junior engineers on clean modular UI architecture and user-centered design standards.

Cisco & Vodafone BMS Dashboards & Technoyana Specialized Engineering

Cisco & Vodafone Dashboards

Building Management Systems (BMS) Unified Interface

- **Facility Dashboard at Cisco:**

Designed and developed an interactive facility dashboard at Cisco Systems campus as a full-time consultant (2009–2010).

- **Unified BMS Control:**

Connected to and visualized diverse Building Management Systems: Air Handling Units (AHUs), Variable Refrigerant Volume (VRVs), industrial chillers, and power meters.

- **Environmental Telemetry:**

Provided facility managers with real-time operational status, thermal mapping, energy consumption, and alarm thresholds.

- **Formal Cisco Recognition:**

Received official Certificate of Appreciation from Cisco leadership in 2010 for outstanding visualization and operational utility.

- **Vodafone Solutions:**

Delivered specialized data visualization dashboards for Vodafone operations to monitor network and facility metrics.

Specialized Engineering Projects

CAD/CAM, CNC, 3D Visualization & Mobile Products

- **Architectural Visualization:**

Delivered multiple architectural 3D visualizations, spatial renderings, and walkthroughs for commercial and real estate projects.

- **CAD/CAM & CNC Development:**

Led customized CNC machine development and automated CAD/CAM toolpath pipelines, bridging physical fabrication with digital software.

- **SellAny Mobile Marketplace:**

Built and launched 'SellAny'—a consumer mobile application allowing users to list and sell items effortlessly.

- **Payment Parking Mobile App:**

Led an engineering team to develop an automated parking payment mobile application with real-time slot occupancy tracking.

- **Flight Simulation Visuals (CAE):**

Visual Database Developer creating 3D terrain and aircraft simulation databases under locked 60 FPS budgets.

Technical & Managerial Mastery Across 18+ Years of Delivery

1. Scalable Frontend & Modular Architecture

- React, Angular, TypeScript, JavaScript, Node.js, Web Components (LitElement).
- Architected Enterprise UI Design Systems used across diverse development teams.
- State management, micro-frontends, REST, WebSockets, and streaming telemetry.
- 4-year zero-downtime migration of Cisco Stadium Vision from Flash/Flex to Angular/React.

2. Spatial 3D, CAD & Generative AI

- Voxelforge AI (generative modular 3D for Unreal/Unity/WebGL) & Ayam3d (parametric 3D).
- Architectural 3D visualization, animated walkthroughs, and CAD/CAM CNC machine development.
- Flight simulator 3D terrain and aircraft databases under locked 60 FPS budgets (CAE).
- Interactive 3D parts inspection explorer supporting millions of catalog components.

3. High Reliability & Operational Security

- Delivered Philips ICU healthcare software meeting stringent clinical safety benchmarks.
- Strict cybersecurity enforcement: Multi-factor authentication (MFA), XSS mitigation, RBAC.
- Offline-first mobile architecture (IndexedDB/Service Workers) with conflict-free cloud sync.
- Building Management System (BMS) unified interface for AHUs, chillers, and telemetry (Cisco).

4. Agile Leadership & Startup Delivery

- Founder/Co-Founder: Technoyana, Twitan.com, SrushtiLabs, InnoBrik.
- Associate Project Manager at UST Global; managed cross-functional squads for Cisco.
- Early user prototyping, design-to-code pipelines, and active stakeholder governance.
- Honored with Cisco Appreciation (2010) and 3 UST Global Excellence Awards (2015, 2016, 2018).

SECTION 3 · CONCLUSION & ROLE ALIGNMENT

Alignment to Senior Manager – IT & Continuous Learning Pledge

✓ Design and Engineering Synthesis:

Holding a postgraduate degree in Human Interface Design (M.S. in Computing, UK) combined with an engineering degree in Electronics and Communication (B.E.), I bring a rare, balanced capability spanning both human-centered UX design and deep technical software architecture.

✓ Proven Breadth Across Startup Velocity & Enterprise Scale:

My 18+ years of experience encompass leading agile, high-velocity startup product incubation (Technoyana, SrushtiLabs, Twitan, InnoBrik) as well as managing complex multi-year enterprise projects with stringent quality, security, and uptime mandates (Philips ICU Healthcare, Cisco Systems, UST Global, Ness).

✓ Handling Products Across Every Stage of the Lifecycle:

Demonstrated expertise across every product phase: early discovery prototypes, building reusable design systems, legacy system migrations (Flash to Angular/React), production deployment, security compliance, and long-term customer support.

✓ Pledge for Self-Training & Certification:

I am fully committed to the success of the organization and team. As per the specific operational requirements of the Senior Manager – IT role, I pledge to proactively upskill myself and complete any required enterprise certifications (e.g., Cloud Architecture, ITIL, Cybersecurity, or project governance) to align completely with team goals and project needs.

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